

BPM REV-B DFE ACCEPTANCE TEST PROCEDURE

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DRAFT

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1. General

1.1. Scope

This procedure verifies the functional operation of the NSLS-II RF BPM Zynq Ultrascale+ DFE board. Data collected during testing supports acceptance decisions and performance statistics.

1.2. Documents

Schematics and assembly drawings pertinent to the procedures:

Item	Document Number	Document Title
1	LT-EL-SR-EU-BPM-0614-03	DFE Schematic

Table 1: Applicable RF BPM Drawings

2. Requirements

2.1. Safety

Personnel performing tests herein should, at a minimum, be qualified in the following:

Item	Course Number	JTA
1	HP-OSH-150A	Basic Electrical Safety
2	TQ-ELECSAF1	Electrical Safety I

Table 2: Safety Qualifications

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2.2. Test Equipment

The following table delineates the instruments used to conduct the measurements described throughout the procedure. Equipment and tools other than those suggested may be supplemented at the discretion of the cognizant engineer (CE), provided they are equivalent and the appropriate parameters are satisfied.

Item	Nomenclature	Description
1	Lab PC	Any general-purpose computing device that runs Ubuntu OS/windows
2	Software	Python 3.3 Vivado 2022.2 Vitis 2022.2 LTPowerPlay (only works on Windows Machine) EPICS (epics base & pscdriver libraries installed)
3	Digital Multi-meter	Fluke 287 True RMS
4	Power Supply	RIGOL DP831A DC Power Supply
5	LTPowerPlay Controller	DC1613A USB to I2C Controller

Table 3: Test Equipment

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3. Test Setup

3.1. AFE Board

- 3.1.1. Connect AFE board to the AFE connectors of the DFE board.
- 3.1.2. Connect the RF signal generator output to a 4-way splitter.
- 3.1.3. Then, the 4 outputs of the splitter will be attached to J1, J7, J11, and J15 of the AFE board.

3.2. Power Supply

- 3.2.1. Connect a 5 V Power supply to J12 of the DFE Board.
- 3.2.2. Connect a 6 V Power supply to J9 of the AFE Board.

3.3. JTAG Cable

- 3.3.1. The Xilinx JTAG connector consists of a power cable, ethernet cable, and JTAG cable.
- 3.3.2. Power on the JTAG connector by plugging the power cable into a power terminal.
- 3.3.3. Then, connect the ethernet cable from the local network to JTAG connector.
- 3.3.4. Finally, connect the JTAG cable to J10 on board.
- 3.3.5. Take note of the IP Address for the JTAG cable.

Tip: Try pinging the IP address to check the connection.

3.4. SFP Connector

- 3.4.1. Take a fiber cable and connect it to the SFP connector on the DFE board.

3.5. Serial Port

- 3.5.1. Connect an ethernet cable from MOXA to the serial port (J2) on board.
- 3.5.2. Take note of the IP Address for the serial port.

3.6. Gigabit Ethernet

- 3.6.1. Connect an ethernet cable from the local network to the ethernet port (P5) on board.

3.7. Git Repositories

- 3.7.1. Clone this git repository into local machine:
 - git clone https://github.com/kbouth/dfc_accept_test.git (HTTPS)

3.8. microSD Card

- 3.8.1. Take an empty microSD card and copy **dfc_accept_test/fpga_boot/BOOT.bin** into it.

3.9. LEMO I/Os

- 3.9.1. Connect LEMO cables to the LEMO I/Os (Figure 3).
 - OUT0 → IN0
 - OUT1 → IN1
 - OUT2 → IN2
 - OUT3 → IN3

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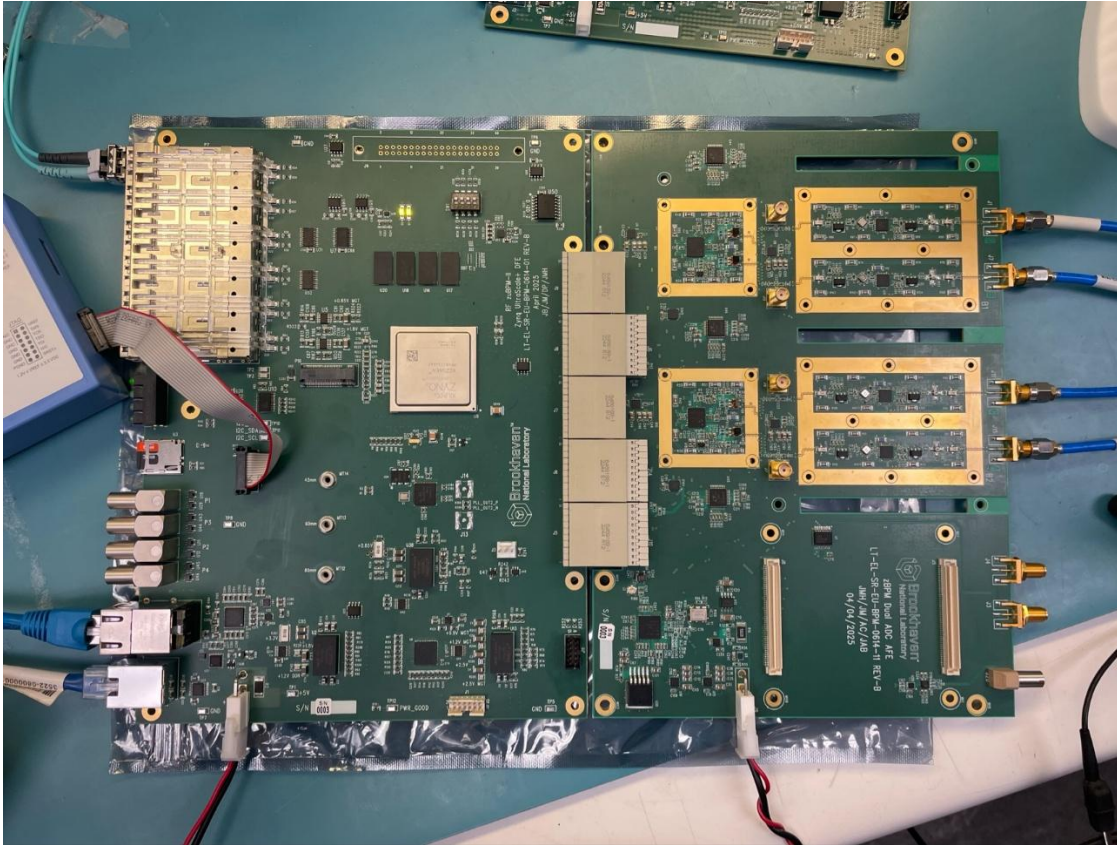


Figure 1: Test Setup.

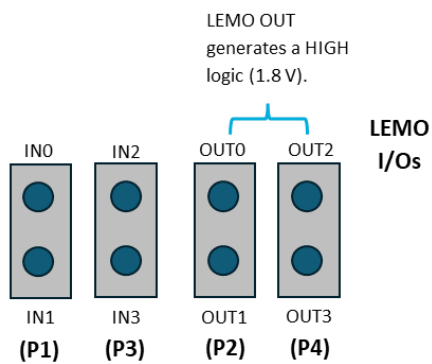


Figure 2: LEMO I/O Diagram.

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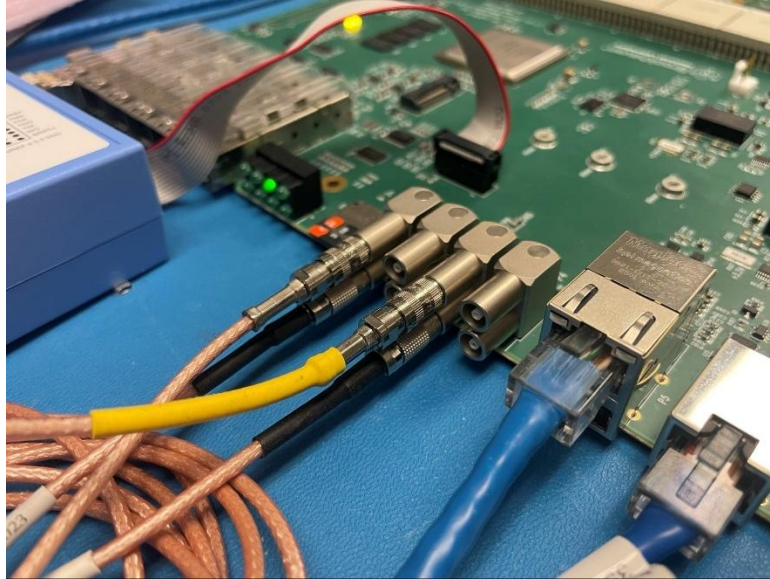


Figure 3: LEMO IN0 connected to OUT0 and IN1 connected to OUT1. For testing, all LEMO outputs should be connected to their respective LEMO input.

3.10. LTC2977 Power Module

- 3.10.1. Supply the DFE board with 5V.
- 3.10.2. If using a non-windows machine, import zudfe_ltc2977_pwr_seq.proj to a windows machine. (LTPowerPlay Software only works on a windows machine).
- 3.10.3. Plug DC1613A Controller to J1.
- 3.10.4. Run LTPowerPlay Software on a **Windows Machine**.
- 3.10.5. On the opening window, open project and select zudfe_ltc2977_pwr_seq.proj.

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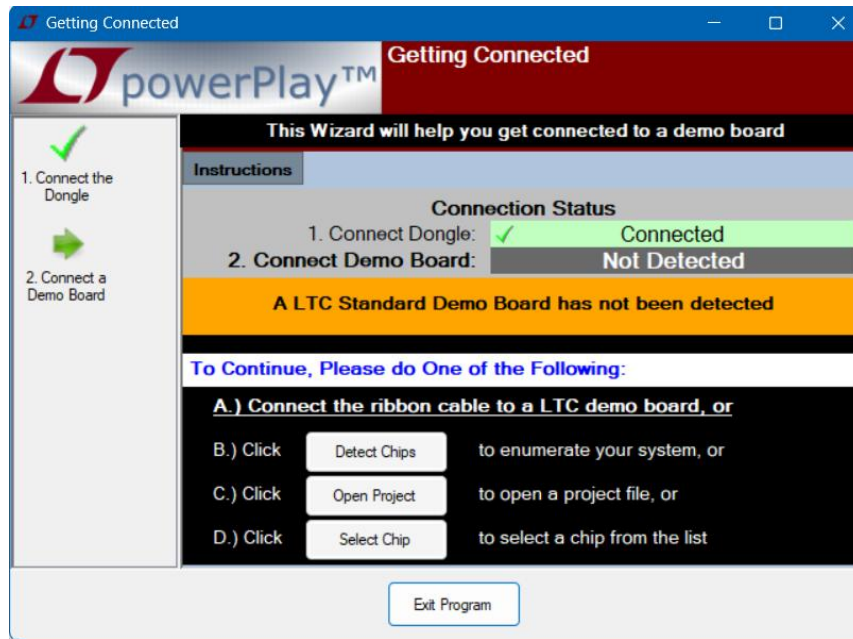


Figure 4: LTPowerPlay Home Menu. The software should autodetect the DC1613 controller when LTPowerPlay opens.

3.10.6. Once the project is open, click on “Go online”.

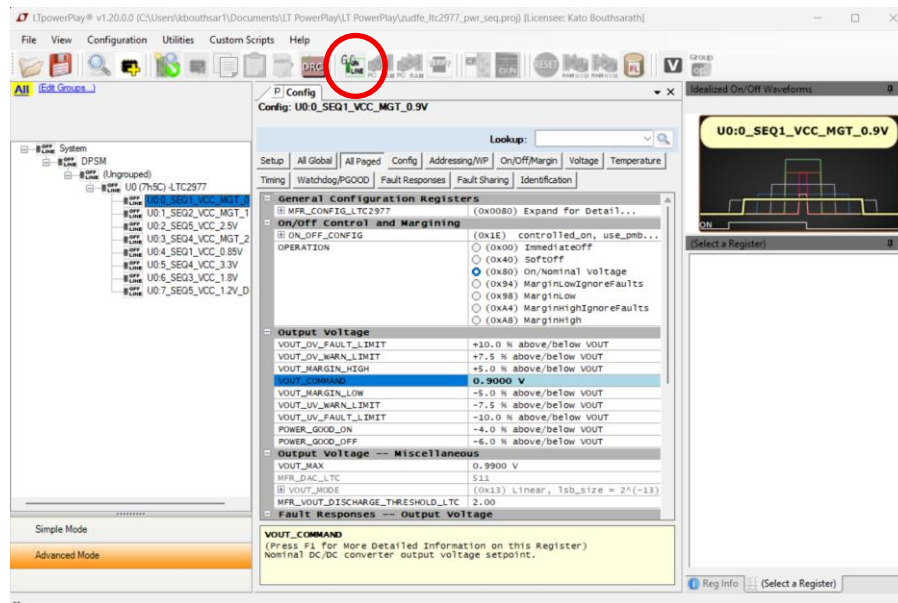


Figure 5: Go Online.

3.10.7. Then, click on “PC to RAM” to write the voltage configuration to memory.

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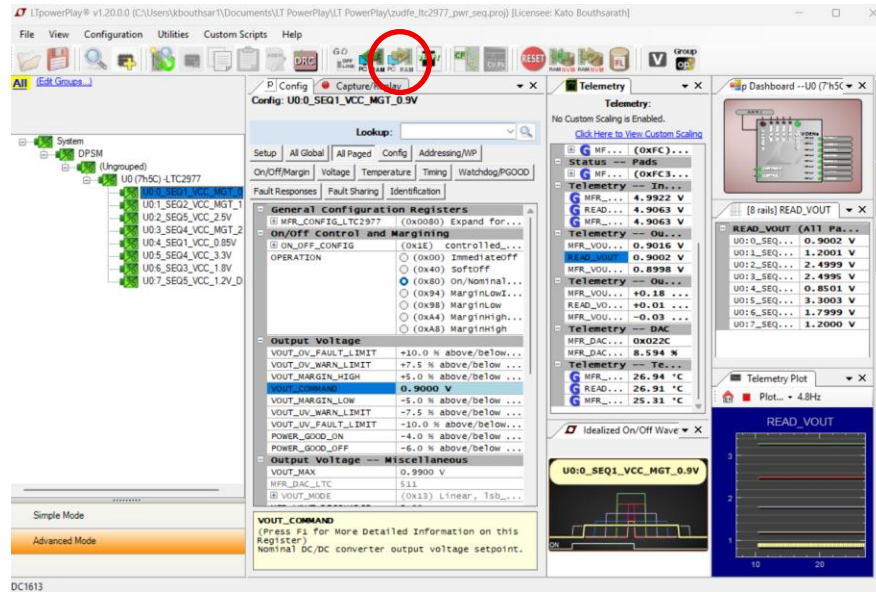


Figure 6: PC to RAM.

3.10.8. Finally, click on RAM to NVM to write the voltage configuration to non-volatile memory.

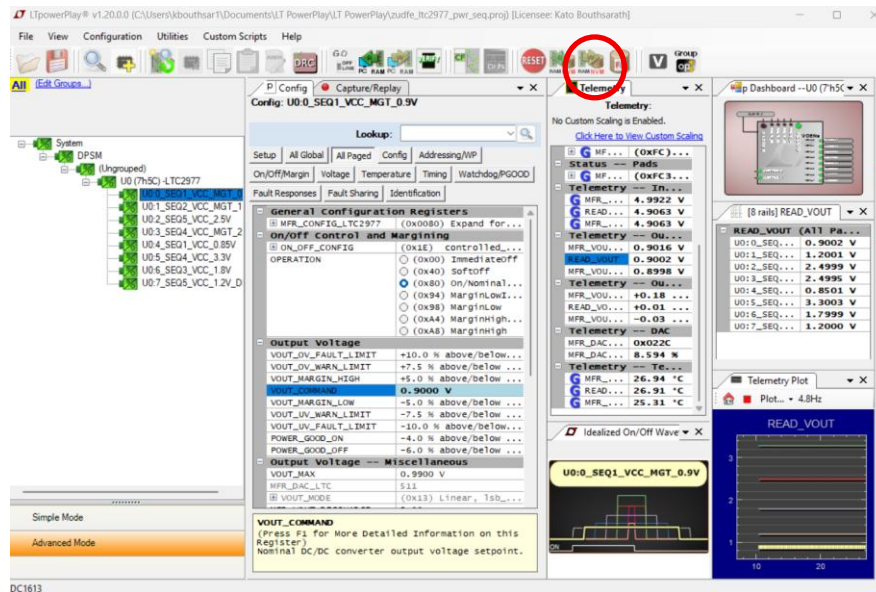


Figure 7: RAM to NVM.

3.10.9. Unplug the controller from the board.

3.10.10. Power-cycle the board and measure these test points (Table 4) to verify that the board is getting the proper voltage. Record the values to input them into the GUI later. Refer to Figure 8 for the locations of the test points of the zudfe.

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Test Point	Voltage
C1	+0.9 V (± 0.05)
C2	+1.2 V (± 0.05)
C3	+2.5 V (± 0.05)
C4	+2.5V (± 0.05)
C211	+0.85 V (± 0.05)
C5	+3.3 V (± 0.05)
C73	+1.8 V (± 0.05)
C74	+1.2 V (± 0.05)

Table 4: DFE Regulators Voltage Measurements

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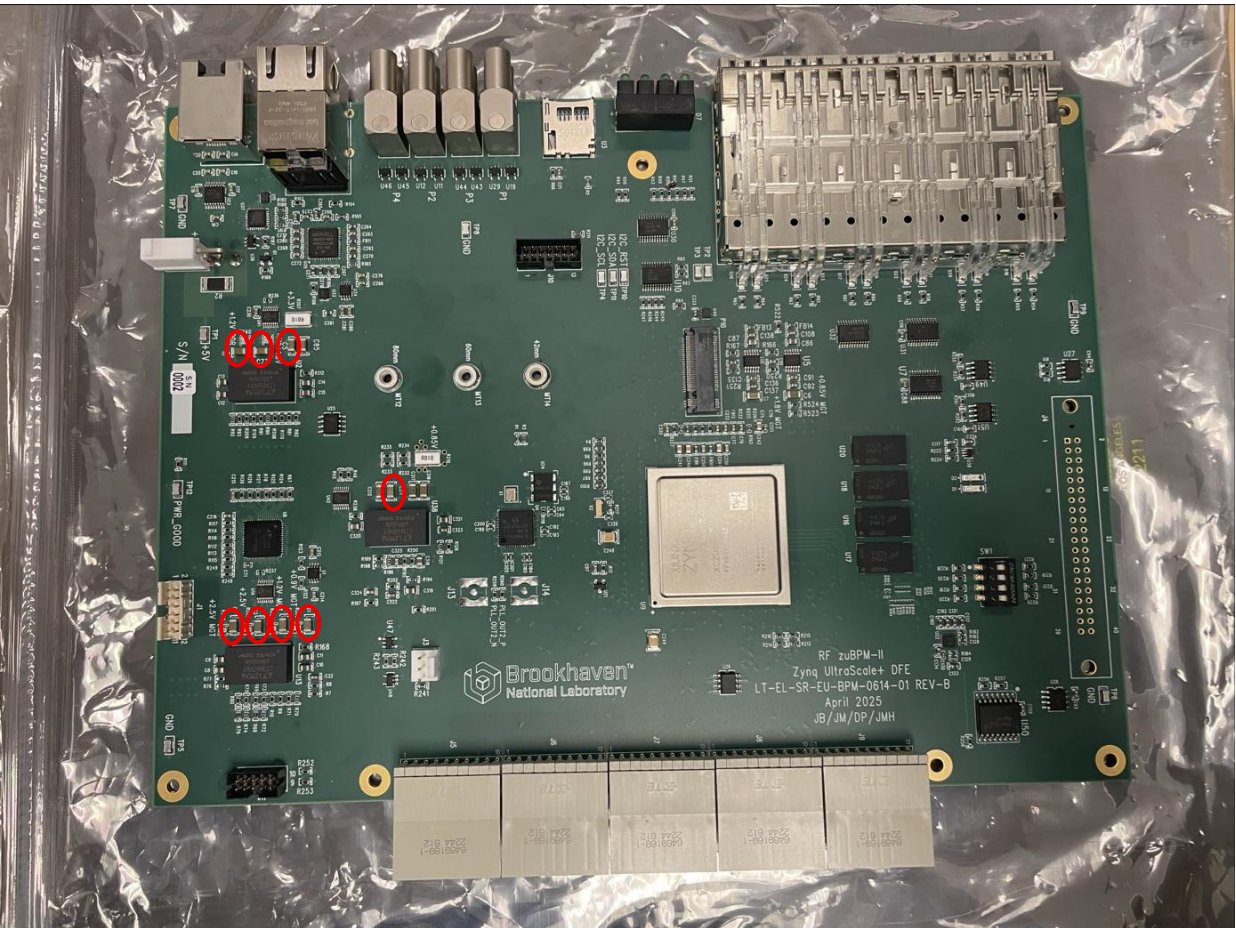


Figure 8: Test Points

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3.11.DFE Python GUI Tutorial

- 3.11.1. Open a terminal and run the python script by typing “python3 dfe_accept_gui.py”
- 3.11.2. This will open the python GUI, which most of the testing will occur.

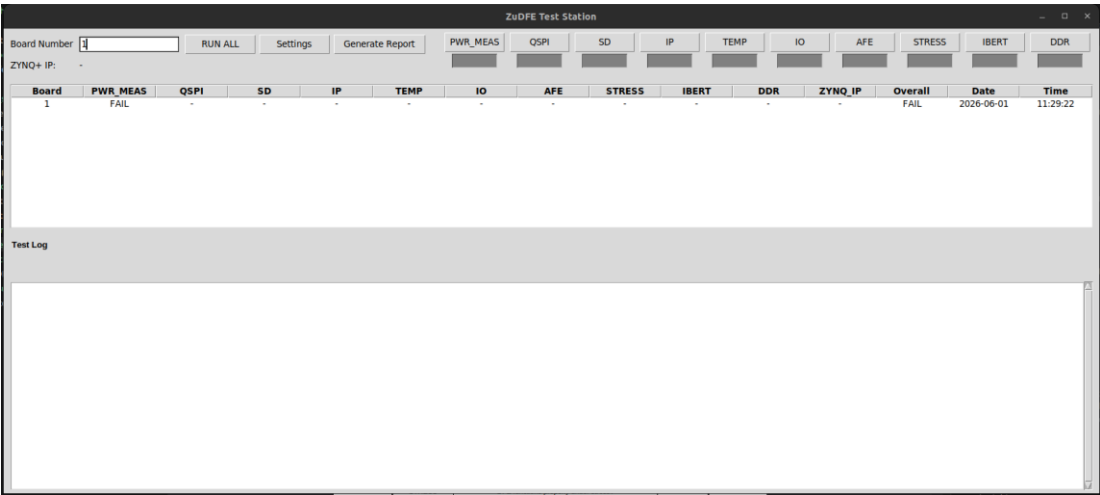


Figure 9: ZuDFE Test Station GUI

- 3.11.3. Click on settings button to change the ip addresses assigned to the JTAG, MOXA serial port, and paths to the Vivado and Vitis executables.
- 3.11.4. The bottom window is the test log for the current board being tested and keeps track of the serial terminal while the tests are being run. The test log is saved externally in the **/dfe_accept_test/logs** directory.
- 3.11.5. If the user wants to generate the report, they must press the “Generate Report” button, and it will generate a PDF report of all the results.
- 3.11.6. The user has multiple options of how they would like to run the test:
 - 3.11.6.1. **RUN ALL:** The user can run through all the tests sequentially. To start, press the RUN ALL button on the GUI.
 - 3.11.6.2. **Individual Tests:** The user can run an individual test of their choice. Press any of the buttons on the GUI to start the test. If it passes, then the LED will turn green. If it fails, then the LED will turn red.

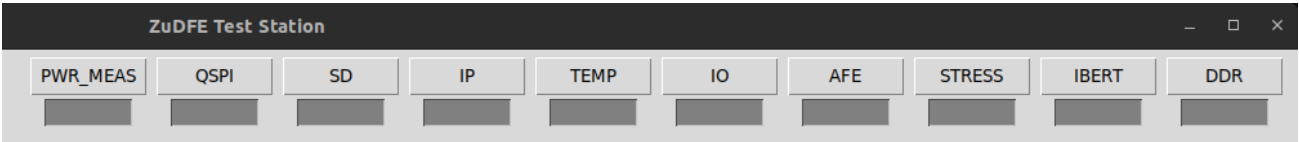


Figure 10: Individual buttons for each test.

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3.11.7. Once a test runs, it will print the output on the test log window. After the completion of the test, it will record the result on the results window.

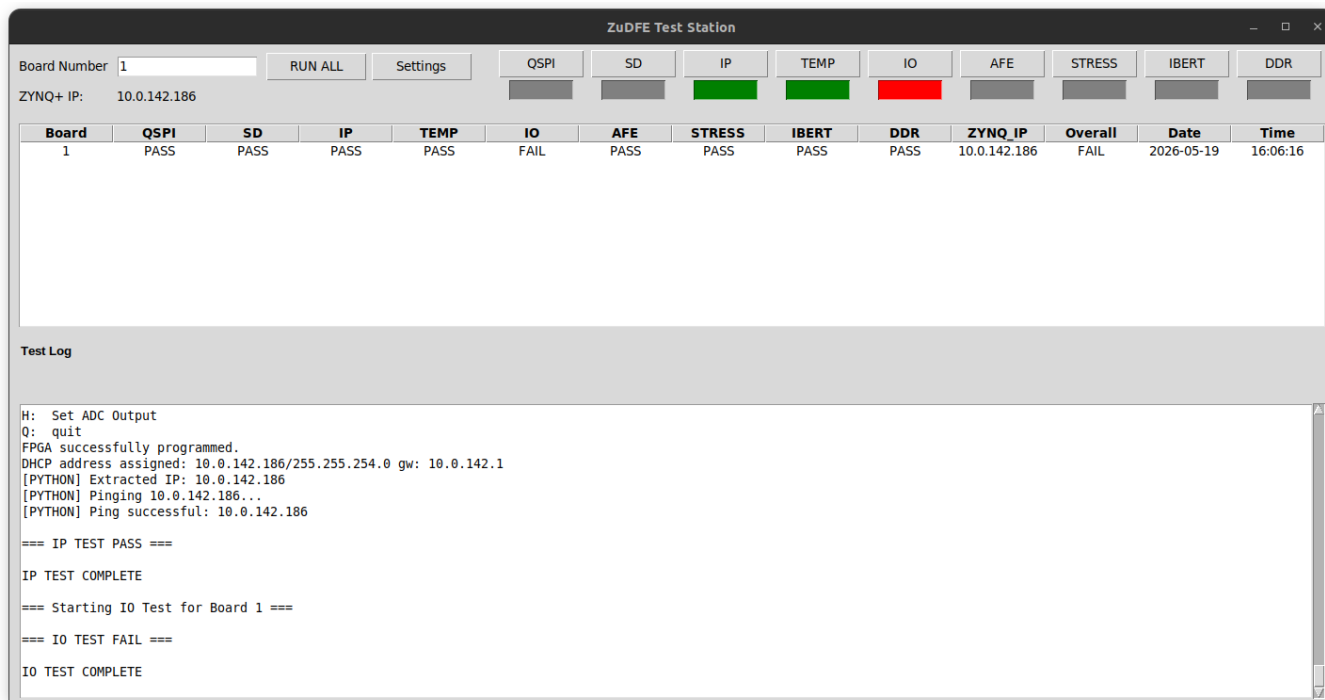


Figure 11: Example of a test being run in the GUI. One test pass and one fails.

4. DFE Verification Tests

This section will describe each test that will be run to verify the functionalities of the DFE boards.

4.1. Power Measurement Test

4.1.1. The power measurement test determines if the DFE board is being powered properly with the proper power rails.

4.1.2. Measure the test points (Table 4) and record the values in the GUI.

4.1.3. The script will compare each measured voltage to their corresponding reference voltage. Then, it'll determine if it passes or fails.

4.2. QSPI Boot Test

4.2.1. The QSPI boot test determines if the FPGA can boot the firmware via SD card once the DFE board is power cycled.

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- 4.2.2. The python script will execute `/dfe_accept_test/fpga_boot/qspi_boot.sh` that'll automatically flash program the QSPI (U50).
- 4.2.3. Once flash programming is complete, the python GUI will instruct the user to:
 - 4.2.3.1. Power off the board.
 - 4.2.3.2. Set BOOT Mode to QSPI (0001) (Figure 12).
 - 4.2.3.3. Power on the board.
- 4.2.4. Then, the python script will read from the serial terminal to determine if the FPGA booted properly via QSPI.

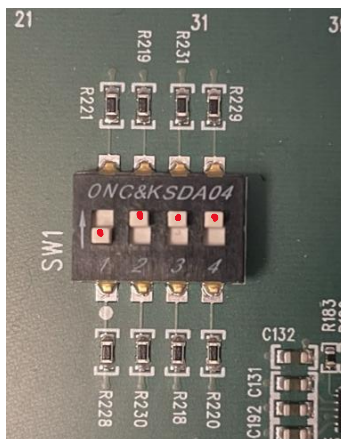


Figure 12: BOOT mode (SW1) set to QSPI.

4.3. SD Boot Test

- 4.3.1. The SD Boot Test determines if the FPGA can boot the firmware via SD card once the DFE board is power cycled.
- 4.3.2. On the GUI, it will instruct the user to:
 - 4.3.2.1. Plug microSD card to the SD slot on the board (U3).
 - 4.3.2.2. Set BOOT mode to SD (0101) (Figure 13).
 - 4.3.2.3. Power cycle the board using the 5V power supply.
- 4.3.3. Then, the python script will be read from the serial terminal to determine if the FPGA booted properly via SD.

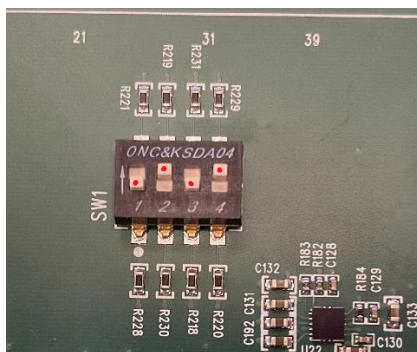


Figure 13: BOOT mode (SW1) set to SD.

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4.4. IP Address Test

- 4.4.1. The IP address test determines if the assigned IP addressed from the DFE can be created and that the user can communicate with it. This IP address is used to establish the EPICS IOC.
- 4.4.2. The test will program the Zynq U+ fpga with the FPGA_boot firmware and print out the output on the serial terminal.
- 4.4.3. Once the fpga is programmed, it should automatically assign an IP address and print it on the serial terminal.
- 4.4.4. Once an IP address is assigned, the python script will try to ping that IP address. If it was able to ping it, the test is a success.

4.5. Temperature Sensor Test

- 4.5.1. The temperature sensor test determines if the temperature sensors on the DFE are within tolerance.
- 4.5.2. The test will program the Zynq U+ fpga with the FPGA_boot firmware and will read the temperatures from the four sensors in the serial terminal.
- 4.5.3. Then, it will verify that the temperatures are within 20-60 degrees.

4.6. LEMO I/O, LEDs, & Debug Header Test

- 4.6.1. This test is to determine if the LEMO I/Os, the LEDs and the debug header work correctly.
- 4.6.2. The test will program the Zynq U+ fpga with the FPGA_boot firmware and print out the output on the serial terminal.
- 4.6.3. The boot firmware configures the LEMO I/Os as shown in figure 2.
- 4.6.4. Pins 17,18,19, and 20 on the debug header (J4) are configured to read the LEMO outputs (OUT0-OUT3).
- 4.6.5. The LEDs (D7) are programmed to turn on one LED at a time at a rate of 1 second.
- 4.6.6. The python GUI will ask the user to:
 - 4.6.6.1. Measure pins 17,18,19, and 20 on the debug header (J4) with a multimeter. Ensure that they are outputting 1.8 V.
 - 4.6.6.2. Confirm that the LEDs are turned on one by one.
 - 4.6.6.3. If all conditions are met, the test is a success.

4.7. AFE Test

- 4.7.1. The AFE test will determine if the ADC bits are toggling on the AFE board.
- 4.7.2. The test will program the Zynq U+ fpga with the FPGA_boot firmware and print out the output on the serial terminal.
- 4.7.3. From the **/zubpm-ioc** directory, open RELEASE.local file and edit the paths to where you have epics base and pscdriver installed.
- 4.7.4. Save changes and move RELEASE.local into the **/zubpm-ioc/configure** directory.
- 4.7.5. Open a terminal from that directory and type **"make"**.
- 4.7.6. Cd into **../iocBoot/ioczbpm** and open st.cmd to edit the file.
- 4.7.7. Edit line 25 to the assigned ZuDFE IP address from section 3.4 and save.
- 4.7.8. The python GUI will instruct the user to:
 - 4.7.8.1. From a terminal, type **"/st.cmd"** to start EPIC IOC.
- 4.7.9. The python script will generate two test patterns (0xAAAA & 0x5555) to the ADC and readback from the EPIC IOC.

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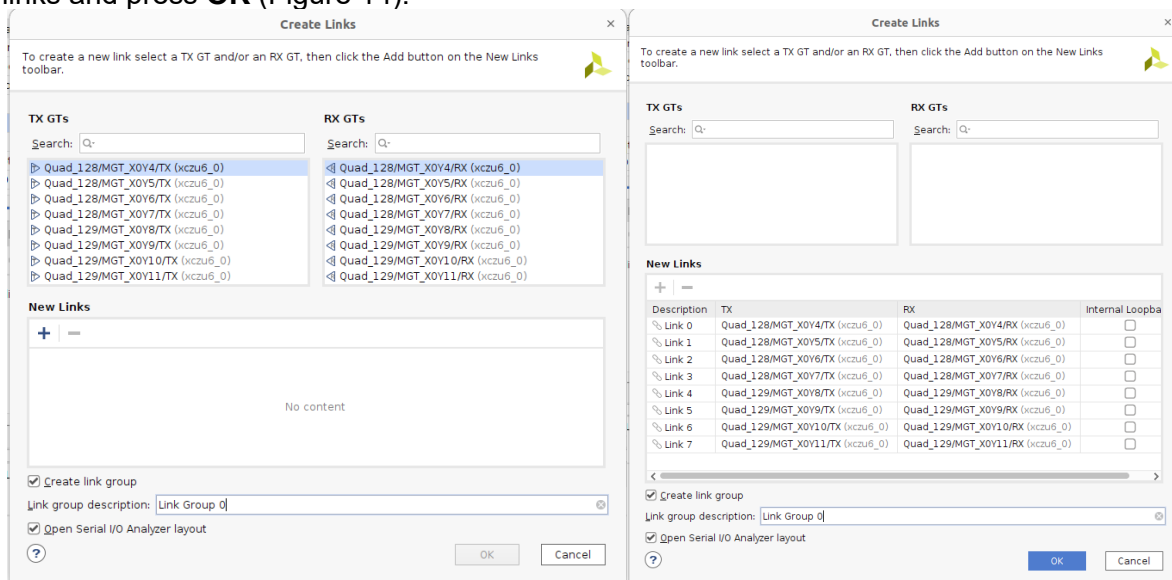
4.8. Stress Test

- 4.8.1. The stress test determines if the FPGA can handle current greater than 5 amps while driving a lot of logic to the FPGA chip.
- 4.8.2. The test will program the Zynq U+ fpga with the FPGA_boot firmware and print out the output on the serial terminal.
- 4.8.3. Once it's programmed, the python GUI will instruct the user to:
 - 4.8.3.1. From a terminal, type **“./st.cmd”** to start EPIC IOC.
- 4.8.4. The python script will confirm if the current for the 0.85 V power rail is greater than 5 amps. If so, then the test is a success.

4.9. GTH Transceivers IBERT Test

- 4.9.1. The IBERT test determines if the six GTH transceivers on the DFE boards work properly.
- 4.9.2. The python script will:
 - 4.9.2.1. Program the FPGA with the ibert_test bit file and debug file
 - 4.9.2.2. Open the Vivado GUI
- 4.9.3. Once the Vivado GUI is open, click on: Open Hardware Manager → Open New Target.
- 4.9.4. On Hardware Server Settings, Connect to: **Remote Server**.
- 4.9.5. Under remote server, set host name to JTAG IP address and click **Next**.
- 4.9.6. FPGA should already be autodetected and listed under Hardware Targets. If FPGA is not listed, you may have to change JTAG clock frequency until it's listed. Then, click **Finish**.
- 4.9.7. If there are no errors, right-click on IBERT under the Hardware menu on the top left. Then, select **Create Links**.

Note: If Vivado claims that there is no debug core programmed, power-cycle the DFE board to boot from SD/QSPI and restart the test. This usually means that the PLL was not programmed properly. Booting the FPGA via SD/QSPI will program the PLL again and allow the user to run the IBERT test again.
- 4.9.8. Under New Links, click on the **+** symbol to add a link between the TX/RX GTs. Add all links and press **OK** (Figure 14).



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Figure 14: Create Links Window. Vivado autodetects any GT transceivers (Left). Users must add each link for IBERT analysis (Right).

4.9.9. Under Serial I/O Links, check that RX PLL Status and TX PLL Status are locked (Figure 15).

4.9.10. Then, change Loopback Mode to “Near-End PMA” for all links.

	BERT Reset	TX Pattern	RX Pattern	TX Pre-Cursor	TX Post-Cursor	TX Diff Swing	DFE Enabled	Inject Error	TX Reset	RX Reset	RX PLL Status	TX PLL Status	Loopback Mode
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None
5E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)	873 mV (11000)	✓	Inject	Reset	Reset	Locked	Locked	None

Figure 15: Serial I/O Links. RX PLL Status & TX PLL Status should be locked to ensure that the clocks are working properly.

4.9.11. Finally, under BERT RESET, click reset for all links. If the status for all the links is 2.5 Gbps (+/- 2%), then the GTH transceivers are working properly (Figure 16).

Name	TX	RX	Status	Bits	Errors	BER	BERT Reset	TX Pattern	RX Pattern	TX Pre-Cursor	TX Post-Cursor
Ungrouped Links (0)											
Link Group 0 (8)							Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 0	Quad_128/MGT_X0Y4/TX (xczu6_0)	Quad_128/MGT_X0Y4/RX (xczu6_0)	2.498 Gbp	4.083E10	0E0	2.449E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 1	Quad_128/MGT_X0Y5/TX (xczu6_0)	Quad_128/MGT_X0Y5/RX (xczu6_0)	2.495 Gbp	4.083E10	0E0	2.449E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 2	Quad_128/MGT_X0Y6/TX (xczu6_0)	Quad_128/MGT_X0Y6/RX (xczu6_0)	2.498 Gbp	4.084E10	0E0	2.449E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 3	Quad_128/MGT_X0Y7/TX (xczu6_0)	Quad_128/MGT_X0Y7/RX (xczu6_0)	2.499 Gbp	4.084E10	0E0	2.449E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 4	Quad_129/MGT_X0Y8/TX (xczu6_0)	Quad_129/MGT_X0Y8/RX (xczu6_0)	2.498 Gbp	4.084E10	0E0	2.448E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 5	Quad_129/MGT_X0Y9/TX (xczu6_0)	Quad_129/MGT_X0Y9/RX (xczu6_0)	2.502 Gbp	4.084E10	0E0	2.448E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 6	Quad_129/MGT_X0Y10/TX (xczu6_0)	Quad_129/MGT_X0Y10/RX (xczu6_0)	2.498 Gbp	4.085E10	0E0	2.448E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)
Link 7	Quad_129/MGT_X0Y11/TX (xczu6_0)	Quad_129/MGT_X0Y11/RX (xczu6_0)	2.498 Gbp	4.085E10	0E0	2.448E-1	Reset	PRBS 7-bit	PRBS 7-bit	0.00 dB (00000)	0.00 dB (00000)

Figure 16: Serial I/O Links. All links should read about 2.5 Gbps.

4.10. DDR Test

4.10.1. The DDR Test determines if the DFE can transfer data at a double data rate or DDR.

4.10.2. The DDR test programs the Zynq U+ FPGA with the sample Xilinx DDR firmware.

4.10.3. Then, it runs two analyses: a read 2D eye diagram analysis and a write 2D eye diagram.

4.10.4. The python script computes the average percentage of the eye opening and determines if it is within 80%.

4.10.5. If both averages are within 80%, then it passes the DDR test.

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PROJECT:

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